Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science and Engineering

Course Code : CSE 3209
Course Title : Digital Signal Processing

Theory Assignment

Submission Date: June 01, 2025

Submitted By:

Eazdan Mostafa Rafin
Roll: 2003055
Section : **A**

Submitted To:

Md. Farukuzzaman Faruk
Assistant Professor
Dept. of CSE, RUET

Discrete Fourier Transform on EEG Signal

Eazdan Mostafa Rafin
Roll: 2003055

June 01,2025

Objective

To perform the Discrete Fourier Transform (DFT) on a segment of an EEG signal, identify the prominent frequency components (specifically alpha and beta waves), and reconstruct the signal using only these components.

Theory

The Discrete Fourier Transform (DFT) is a fundamental tool in signal processing used to analyze the frequency content of discrete-time signals. It transforms a signal from the time domain into the frequency domain, allowing us to identify dominant frequency components.

Given a discrete signal $x[n]$ of length N, the DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi kn/N}, \quad k = 0, 1, \ldots, N-1$$

Here, $X[k]$ represents the amplitude and phase of the frequency component at index k.

The original time-domain signal can be recovered using the inverse DFT:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \cdot e^{j2\pi kn/N}$$

In this experiment, we apply the DFT to an EEG signal to extract and analyze the alpha (8–13 Hz) and beta (13–30 Hz) frequency bands. By performing an inverse DFT on the filtered components, we reconstruct the signal and observe its key rhythmic patterns.

Electroencephalogram (EEG) signals represent electrical activity in the brain and are typically composed of multiple frequency bands:

- **Delta** (0.5–4 Hz)
- **Theta** (4–8 Hz)
- **Alpha** (8–13 Hz)

- **Beta** (13–30 Hz)
- **Gamma** (30+ Hz)

Mathematical Foundation

The Discrete Fourier Transform (DFT) for a signal $x[n]$ of length N is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi kn/N}, \quad k = 0, 1, ..., N-1$$

The Inverse Discrete Fourier Transform (IDFT) is given by:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \cdot e^{j2\pi kn/N}$$

These transformations allow conversion between time and frequency domains.

Implementation

To analyze the EEG signal, we began by loading the dataset from a CSV file and focusing on the "Fp1" channel, one of the 32 available channels. We visualized this signal in the time domain to observe its raw waveform.

Next, we applied the Fast Fourier Transform (FFT) to convert the signal into the frequency domain, revealing the various frequency components present. From this spectrum, we filtered out all frequencies except those within the alpha (8–13 Hz) and beta (13–30 Hz) bands, which are known to be most relevant for brain activity analysis.

After filtering, we used the Inverse FFT (IFFT) to reconstruct the signal using only these components. Finally, we plotted and compared the original and reconstructed signals to observe how well the dominant frequency bands captured the essence of the EEG signal.

Code Example

Listing 1: DFT on an EEG Signal

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq, ifft
# Load dataset and select Fp1 channel
df = pd.read_csv("features_raw.csv")
signal = df["Fp1"]
# Plot time-domain signal
```

```
plt.plot(signal)
plt.show()

fs = 128
N = 1024
eeg_signal = signal[:N]
t = np.arange(N) / fs
# DFT
yf = fft(eeg_signal)
xf = fftfreq(N, 1/fs)
# Plot frequency spectrum
plt.plot(xf[:N//2], np.abs(yf[:N//2]))
plt.show()

# Filter only alpha and beta frequencies (8-30 Hz)
yf_filtered = yf.copy()
for i, freq in enumerate(xf):
    if not (8 <= abs(freq) <= 30):
        yf_filtered[i] = 0
# IFFT to reconstruct signal
reconstructed_signal = ifft(yf_filtered)
# Plot original and reconstructed signals
plt.subplot(2,1,1)
plt.plot(t, eeg_signal)
plt.title("Original EEG Signal")
plt.subplot(2,1,2)
plt.plot(t, reconstructed_signal.real)
plt.title("Reconstructed EEG Signal (Alpha + Beta)")
plt.show()
```

Results and Observations

The frequency spectrum revealed significant activity in the alpha (8–13 Hz) and beta (13–30 Hz) bands. After filtering and reconstructing the signal using only these components, the resulting waveform retained the core shape of the original but appeared smoother and less noisy. This confirms the dominance of alpha and beta waves in the analyzed EEG segment.

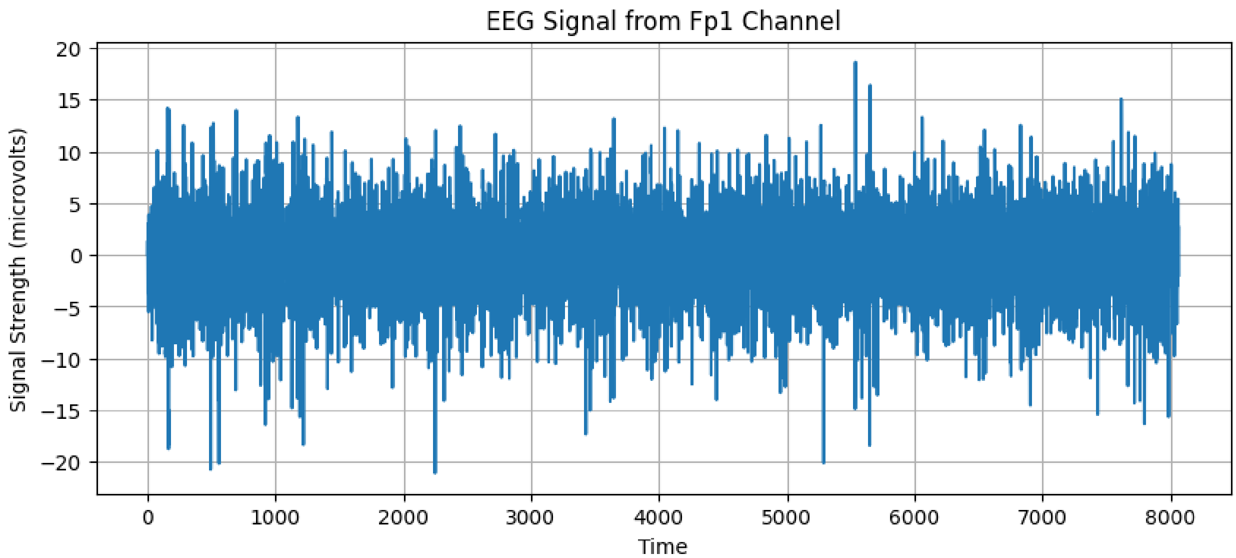

Figure 1: Original EEG Signal from Fp1 Channel

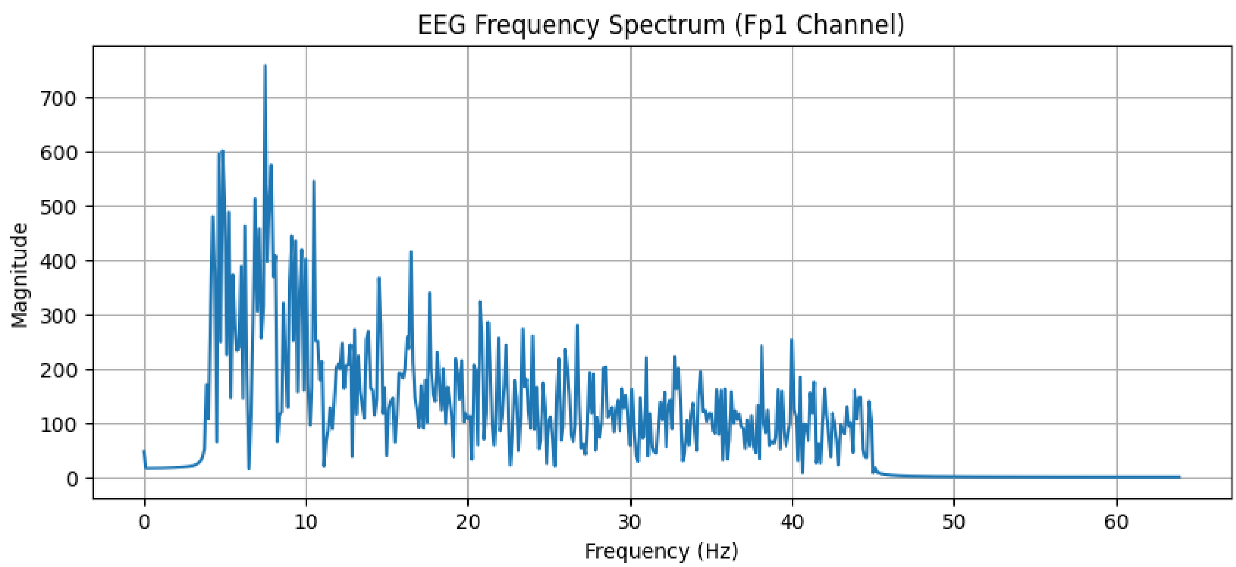

Figure 2: EEG Frequency Spectrum (Fp1 Channel)

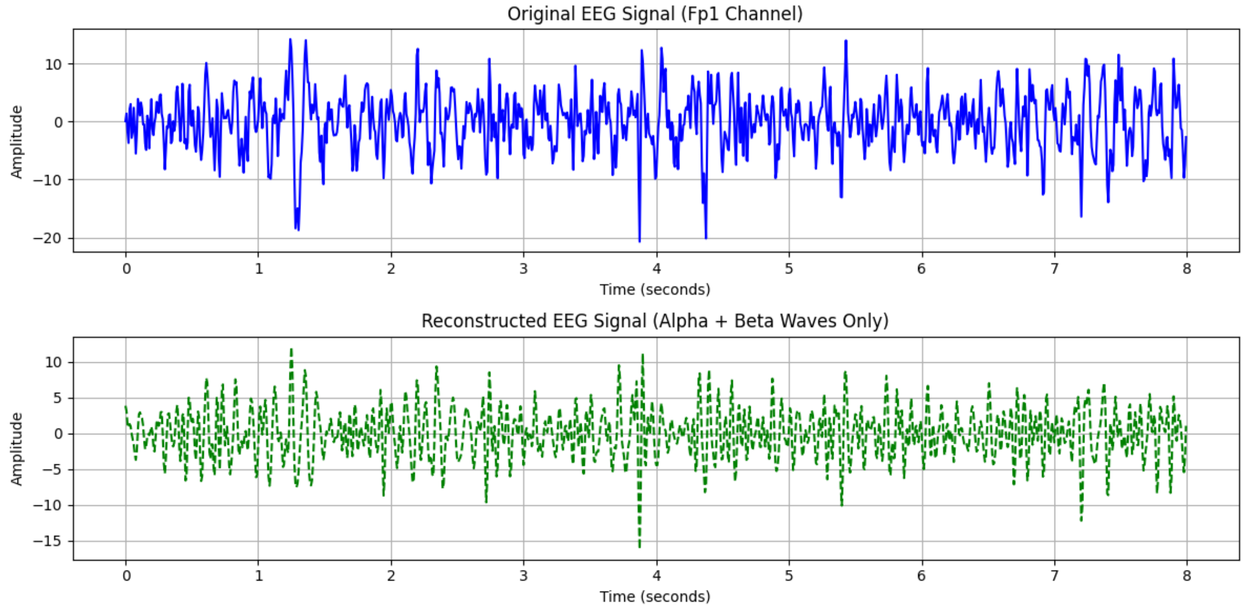

Figure 3: Reconstructed Signal using Alpha and Beta Waves

From the frequency spectrum, we observed prominent peaks in the 8–13 Hz (alpha) and 13–30 Hz (beta) range. When reconstructing the signal using only these components, the overall shape of the signal was retained but with smoother characteristics, indicating successful filtering and reconstruction.

Conclusion

This experiment demonstrated how the Discrete Fourier Transform can be used to analyze EEG signals and isolate specific frequency bands. By filtering out alpha and beta components and reconstructing the signal, we effectively highlighted the brain's rhythmic activities associated with these frequencies. This technique is valuable for EEG analysis, signal denoising, and brain-computer interface applications.